

Ideal gas model

Learning outcomes

By the end of this section you should be able to recognise the key assumptions in the ideal gas model.



Figure 1. A bottle filled with gas can be compressed.

Credit: Proximo, Getty Images

(<https://www.gettyimages.com/detail/photo/crushed-plastic-bottle-royalty-free-image/1302610076>)

Have you ever wondered why air can be compressed, but not ice? When you think of the main difference between these two substances, you may realise that air is a gas, and ice is a solid. Air consists of a mixture of gases, mainly nitrogen gas, N_2 , and oxygen gas, O_2 . How is the behaviour of particles different in gases when compared to solids and how can it help explain the fact that gases can be compressed but not solids?

Practical skills

Inquiry 1: Exploring and designing — Controlling variables

Inquiry 3: Concluding and evaluating — Evaluating

Give the following a try:

- Gather three plastic water bottles.
- Keep the first one empty and close it.
- Fill the second one with water and close it.
- Fill the third one with water, close it and place it in the freezer overnight.

Try squeezing the three bottles. Which can be compressed the most? The least? Why? How does this help explain the arrangement of particles in solids, liquids, and gases?

In section S1.1.2a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/>), we looked at the properties of the different states of matter (solids, liquids and gases). Here, we will look in more detail at the properties of gases, specifically the kinetic molecular theory of gases, which outlines the assumptions of the ideal gas model:

- The particles in a gas are in constant, random, straight-line motion.
- There are negligible forces of attraction (intermolecular forces) between the particles.
- Collisions between particles or with the walls of the container are perfectly elastic (no energy is lost).
- The distance between the particles is much greater than the size of the particles, therefore, gas particles have negligible volume.
- The average kinetic energy of the particles in a gas is directly proportional to the absolute temperature (in kelvin). This means that if an ideal gas at a temperature of 100 K is heated and its temperature doubled to 200 K, then it is assumed that the kinetic energy of the gas particles will also double. If the temperature is halved, then the kinetic energy is halved, etc.

As a gas has no defined shape or volume, it must be contained within a container. The quantity of gas inside the container will be influenced by several variables – volume, temperature and pressure. Volume is often measured in millilitres, expressed as mL or cm^3 , or litres, expressed as L or dm^3 . Temperature is measured by the absolute temperature in units of kelvin. Pressure is determined by how frequently the particles hit the walls of the container and is expressed in Pa or kPa (**Figure 2**).

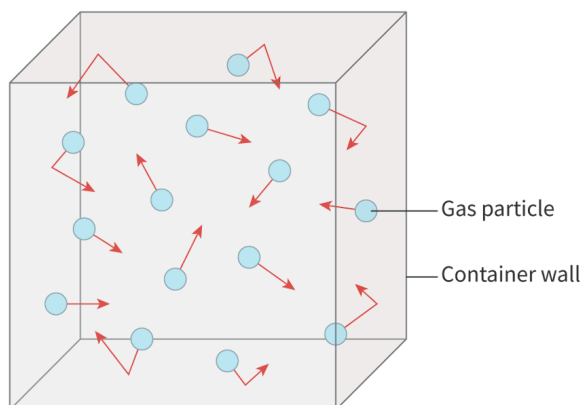


Figure 2. The pressure of the air in a container depends on the frequency with which the particles hit the wall — it is independent of the nature of the particles.

© More information for figure 2

The model proposed by the kinetic molecular theory suggests that, in an ideal gas, the particles are very far apart and moving randomly. In a sample of a gas most of what is there is actually empty space. This predominance of empty space in gas results in the assumption that the size of the gas particle and the interaction of the particles are negligible. The volume of these gases under the same conditions will be the same. Therefore, you can have samples of gas with the same volume, but a different number of gas particles inside because of the empty space. Imagine a school with three classrooms identical in size to the one you are sitting in right

The ideal gas model also proposes that the attractions between molecules can be assumed to be non-existent, and any collisions between them can be assumed to be elastic. In the analogy above, all of these students can move around freely and if they hit the wall or another student, they just bounce off, without losing any energy or getting tired.

Strand: Service

- Demonstrate how to initiate and plan a CAS experience.
- Demonstrate engagement with issues of global significance.

How can you as an IB student help reduce the amount of air pollution produced by your school or in your local community? One possible way is to conduct an energy audit at your school and make a list of suggestions of the ways in which the amount of air pollution produced by your school can be reduced. You can download the [guidance sheet](https://19january2017snapshot.epa.gov/sites/production/files/2016-07/documents/wheel_studentact3.pdf)  (https://19january2017snapshot.epa.gov/sites/production/files/2016-07/documents/wheel_studentact3.pdf) from the Environmental Protection Agency (EPA) for ideas on how to conduct an energy audit of your school.

Real gases

Learning outcomes

By the end of this section you should be able to explain the limitations of the ideal gas model.

We have seen in the previous section that ideal gases are assumed to consist of particles that have negligible volume and negligible attractive forces. Under certain conditions, this can be quite close to reality. However, is this true under all conditions? Can the assumptions still hold when the gas is cooled down or when the gas is pressurised?

All gases deviate from ideal behaviour to some extent and are described as real gases. With real gases, some of the assumptions made about ideal gases no longer apply under certain conditions.

Nature of Science

Aspect: Patterns and trends

The gas laws apply only to ideal gases. How appropriate is it to apply such laws to real gases? Under what conditions do the variations between real and ideal gases become significant?

At low pressure, gas particles are far apart and the forces of attraction between them are negligible, so the gas acts as an ideal gas. At very high pressure the gas particles are closer together and are influenced by forces of attraction. As such, the assumption made in the ideal gas model no longer holds, and the gas cannot be described as an ideal gas anymore. At high pressures, the gas is no longer behaving as an ideal gas and is called a real gas.

At low temperatures, the particles move less rapidly (have lower average kinetic energy). This means that there is a greater opportunity for intermolecular forces between the particles to have an effect. As such, the intermolecular forces of attraction cannot be assumed to be negligible, and the assumption made in the

ideal gas model no longer holds. The gas cannot be described as an ideal gas anymore. At low temperatures, the gas is no longer behaving as an ideal gas and is called a real gas.

In fact, high pressures and low temperatures are precisely the conditions under which gases can be liquified. The ideal gas model can no longer be used to describe the behaviour of the particles in a gas.

Study skills

Real gases deviate from ideal gas behaviour under two conditions: at high pressures and low temperatures.

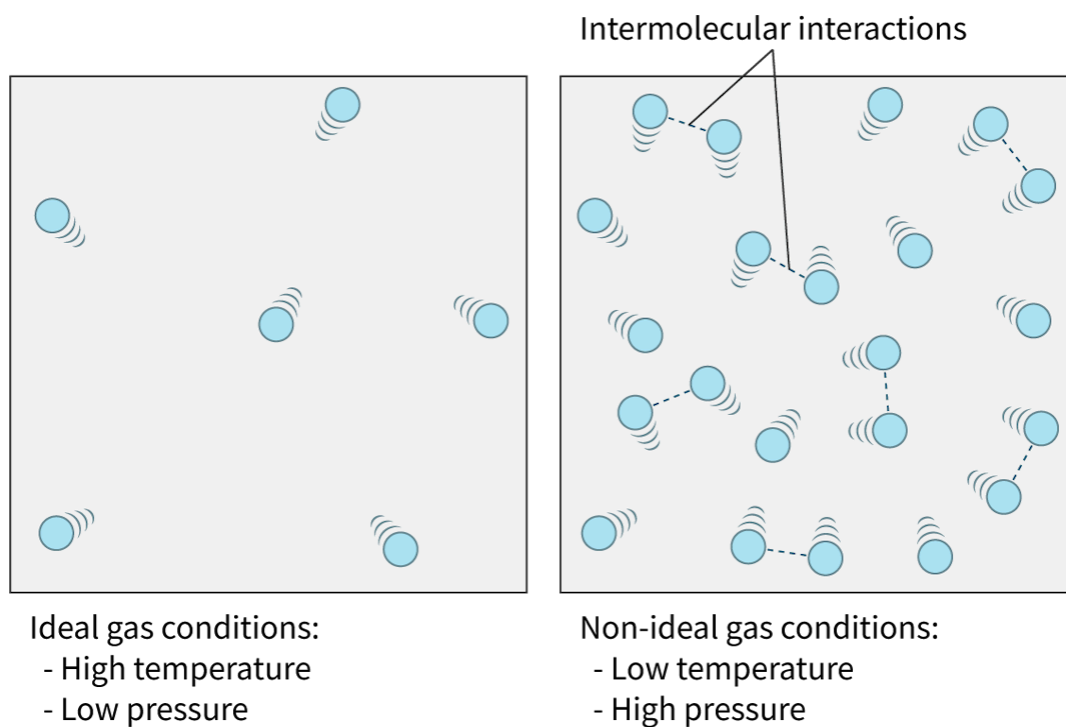


Figure 1. A comparison of the behaviour of a real and an ideal gas.

 More information for figure 1

The image displays two side-by-side diagrams comparing the behavior of an ideal gas and a real gas. On the left, the ideal gas is represented by uniformly distributed particles, illustrated as blue circles with evenly spaced motion lines indicating random movement and no interactions between them. On the right, the real gas is depicted with more densely packed particles, also shown as blue circles, but with dashed lines between them indicating interactions or collisions. The real gas particles show movement lines, suggesting more complex, less uniform motion compared to the ideal gas.

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Would all gases deviate from ideal gas behaviour in the same way?

Making connections

Section S2.2.8 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/intermolecular-forces-id-44359/>). — Under comparable conditions, why do some gases deviate more from ideal behaviour than others?

Since all substances contain different atoms and bonds, they also vary in their intermolecular forces of attraction. Similar substances have the same type of intermolecular forces of attraction. This will be covered more in depth later in the course, but for now, you need to know that for similar substances having the same type of intermolecular forces of attraction, the strength of these attractions will increase as molar mass increases. Which gases will behave more like ideal gases? The ones with smaller or larger molar mass? Think about people in a crowded room: which persons would be able to move through with greatest speed? The smallest people will be able to move through with greatest speed, avoiding collisions with other people. This is true for gas particles as well: the ones that are 'smaller' or lighter are the ones that will move faster. Therefore, gases that show the most ideal behaviour are those with a low molar mass and the weakest intermolecular forces between the particles. For example, among the noble gases, helium has the lowest molar mass and the weakest intermolecular forces of attraction. As such, its behaviour will be closest to the ideal gas behaviour from among the noble gases, since in an ideal gas, the intermolecular forces of attraction are negligible.

Theory of Knowledge

Helium behaves as an ideal gas under a range of temperature and pressure conditions due to its weak intermolecular forces. In addition to filling balloons, helium is used in many industrial applications including medical imaging, semiconductor chips in computers and air bags. Although highly abundant in space, helium is produced by only eight countries in the world. Helium is typically captured as a by-product during the processing of natural gas and once it enters the air, it is nearly impossible to recapture — because it behaves so well as an ideal gas. The global abundance of helium has been declining since 2006, causing massive shortages resulting in delays in manufacturing and medical

treatment worldwide. To what extent should governments be held responsible for the worldwide production and consumption of helium in the face of a shortage?

Activity

- **IB learner profile attribute:**
 - Open-minded
 - Inquirers
- **Approaches to learning:**
 - Thinking skills
 - Being curious about the natural world
 - Asking questions
 - Designing procedures and models
 - Applying key ideas and facts in new contexts
 - Engaging with and designing linking questions
 - Social skills — Actively seeking and considering the perspective of others
- **Time required to complete activity:** 30 minutes
- **Activity type:** Pairs activity

In pairs, you will be making a Brownian motion generator box to observe the effect of increasing pressure, volume and temperature on the behaviour of the gas particles.

1. Find a small cardboard box — of half the size of a shoe box.
2. Place some marbles in the box.
3. Shake the box. Observe how the marbles behave. Do you think this model describes the behaviour of particles in an ideal gas? Discuss with your partner.
4. Increase the pressure by decreasing volume by adding a divider or by adding more marbles. How does the increase in pressure affect the gas particles? Do they behave like an ideal gas more or less with the increase in pressure? Discuss with your partner.
5. Temperature can be decreased by shaking the box less, since temperature affects the kinetic energy of the particles. How does the decrease in temperature affect the gas particles? Do they behave like an ideal gas more or less with the decrease in temperature? Discuss with your partner.

You should be able to see how the particles will then be closer to one another, therefore having less space between them and allowing for more interactions. This creates a larger deviation from ideal gas behaviour.

Molar volume of an ideal gas and gas laws

Learning outcomes

By the end of this section you should be able to investigate the relationship between temperature, pressure and volume for a fixed mass of an ideal gas and analyse graphs relating these variables.

During hot days, bicycle and car tyres become inflated, increasing the volume of the air inside the tyre. Increasing the temperature causes the particles to move with greater speed, resulting in more frequent collisions with the wall of the tyre, expanding the volume. On colder winter days, the volume of the air inside the tyres decreases. The gas particles then move more slowly, having less frequent collisions with the wall of the tyre. This is often why car or bike tyres need to be inflated or deflated a bit. This is just one example of the relationship between variables describing gases, here between volume and temperature.

Molar volume

Recall from [section](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/>) S1.5.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/real-gases-id-44351/>) that gases behave ideally under low pressure and high temperature, allowing the particles to move freely without influence by the other gas particles. Since temperature and pressure conditions vary from place to place, affecting the behaviour of a gas, scientists often refer to standard conditions to describe the behaviour of gases so that their experiments can be reproduced. One set of standard conditions is called STP – standard temperature and pressure – where the temperature is 273 K (0 °C) and the pressure is 100 kPa.

Under standard conditions of temperature and pressure (STP), the volume of 1 mole of any gas will always occupy 22.7 dm³. This value is called the molar volume (volume of 1 mole) of a gas at STP. This can be used to convert between volume and moles of gas at STP. The formula that can be used is:

$$\text{Amount (in moles)} = \frac{\text{volume (dm}^3\text{)}}{\text{molar volume (dm}^3\text{ mol}^{-1}\text{)}}$$

or, at STP:

$$\text{Amount (in moles)} = \frac{\text{volume (dm}^3\text{)}}{22.7 \text{ (dm}^3\text{ mol}^{-1}\text{)}}$$

Study skills

The molar volume of a gas at STP is $22.7 \text{ dm}^3 \text{ mol}^{-1}$. Conditions of STP are a temperature of 273 K and a pressure of 100 kPa. This information can be found in [section 4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/unit-conversions-and-standard-conditions-id-45106/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/unit-conversions-and-standard-conditions-id-45106/) of the DP chemistry data booklet.

Be careful not to confuse the conditions of standard temperature and pressure (STP) with those of standard conditions which you will come across in thermochemistry sections. Standard conditions in thermochemistry include a temperature of 298 K.

In [section 4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/unit-conversions-and-standard-conditions-id-45106/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/unit-conversions-and-standard-conditions-id-45106/) of the data booklet conditions are given for the SATP, which is the standard ambient temperature and pressure. The temperature at SATP is 298 K rather than 273 K.

Worked example 1

Determine the volume occupied by 0.500 mol of chlorine gas (Cl_2) at STP.

Solution steps	Calculation
Step 1: Write out the values given in the question.	$n = 0.500$ at STP 22.7
Step 2: Write out the equation.	$V =$
Step 3: Substitute the values given.	$V =$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$V =$

Worked example 2

Determine the amount (in mol) of helium gas in a balloon of volume of 33.0 dm^3 at STP.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$V = 33.0 \text{ dm}^3$ at STP, molar v
Step 2: Write out the equation.	$\text{mol } (n) = \frac{\text{---}}{\text{molar volume}}$
Step 3: Substitute the values given.	$n = \frac{33.0}{22.7}$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$n = 1.45 \text{ mol}$

Recall from [section S1.4.6 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/), when you learned about Avogadro's law – that equal volumes of gases at the same temperature and pressure contain equal numbers of particles. What happens when we change variables affecting gases? In this section, we will explore relationships between various variables describing gases, namely volume and temperature, volume and pressure, volume and moles, and pressure and temperature. These relationships are described by the gas laws (Boyle's law, Charles' law, and Gay-Lussac's law). The general trends in behaviour with changing conditions of temperature and pressure are the same for all gases. In each case one of these parameters is kept constant, while the relationship between the other two is explored.

Study skills

The names of specific gas laws will not be assessed but you will need to be able to describe the behaviour of a gas when subjected to changes in temperature, volume and pressure.

Relationship between volume and pressure

Boyle's law named after the Irish scientist Robert Boyle (1627–1691), states that at constant temperature the pressure and volume of a fixed mass of an ideal gas are inversely proportional to each other (**Figure 1**). So, if the pressure of a fixed mass of gas at constant temperature is doubled, the volume halves. This can be seen when the plunger of a closed syringe is pushed: as more pressure is exerted, the volume of the enclosed air decreases.

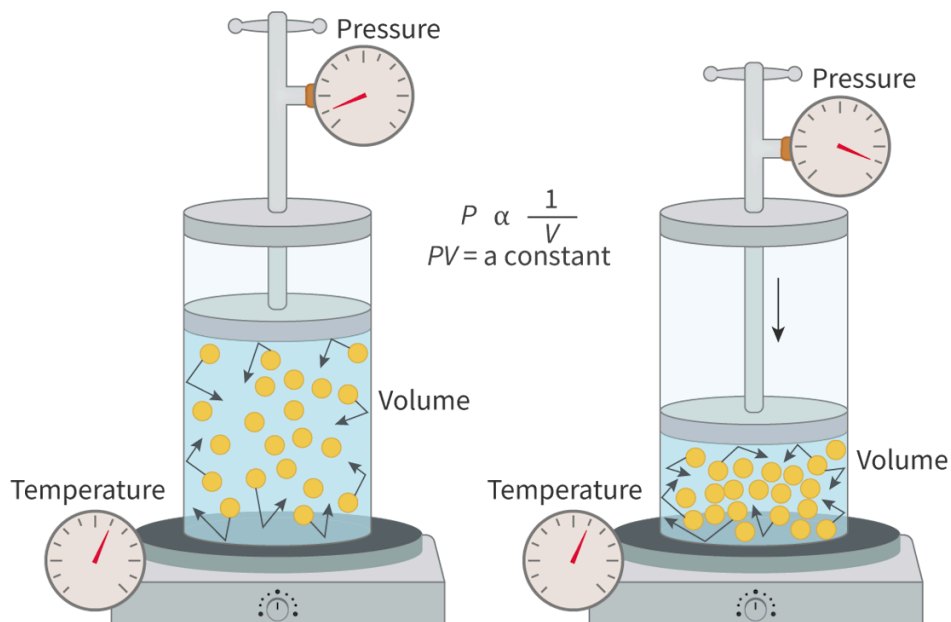


Figure 1. As the pressure on a gas is doubled, its volume is halved.

 More information for figure 1

The image shows two cylindrical containers side by side, each with a movable piston at the top and both under constant temperature. The left container has a lower pressure gauge reading and a larger volume, while the right container has a higher pressure gauge reading and a smaller volume. Yellow particles, representing gas molecules, are inside both containers exhibiting random motion. This demonstrates Boyle's Law, where pressure and volume of a gas are inversely proportional when temperature is constant. Arrows indicate the movement of gas particles. A formula is noted between the containers indicating the inverse relationship: ($P \propto \frac{1}{V}$) and ($PV = \text{a constant}$).

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This relationship can be visualised graphically as in **Figure 2**. Look carefully at the plotted variables below. What do you notice?

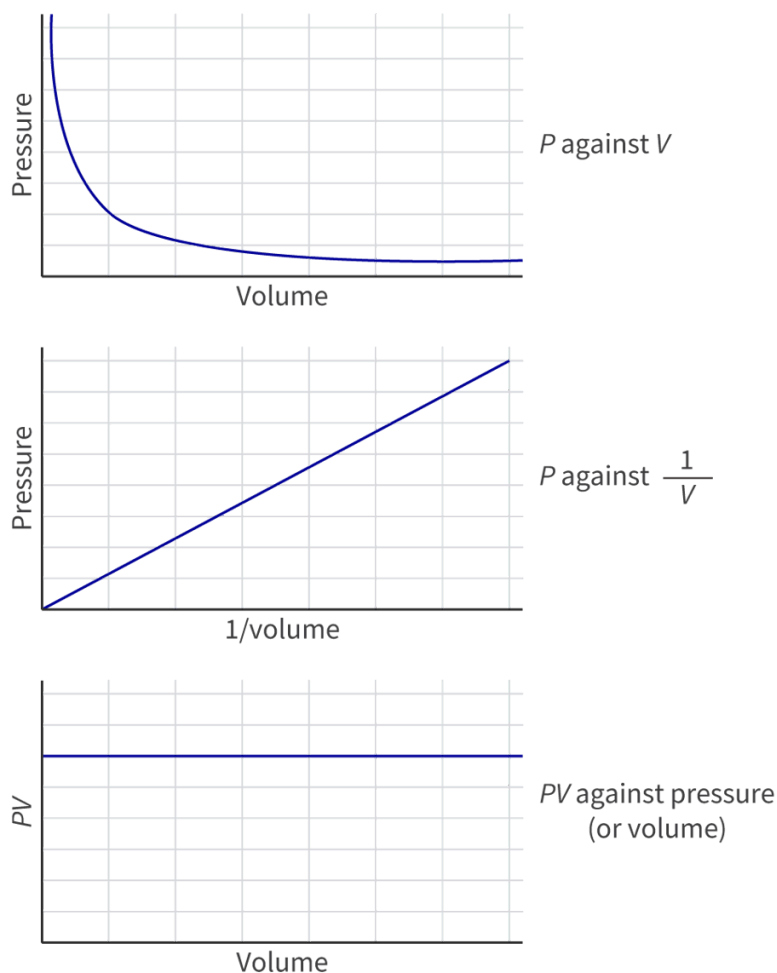


Figure 2. Boyle's law graphical illustration.

 More information for figure 2

The image contains three graphs illustrating Boyle's law relationships.

1. The first graph shows "Pressure" on the y-axis versus "Volume" on the x-axis. The curve illustrates an inverse relationship, where the pressure decreases as the volume increases, depicted by a downward sloping curve.
2. The second graph represents "Pressure" on the y-axis against "1/Volume" on the x-axis. In this graph, a linear positive correlation is shown with a straight line moving diagonally upward, indicating that pressure is directly proportional to the inverse of volume.
3. The third graph displays "P*V" on the y-axis, captioned as "PV against pressure (or volume)". This graph shows a horizontal line, suggesting that the product of pressure and volume remains constant regardless of changes in individual variables.

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As observed in the above graphs, pressure and volume are inversely proportional

$$\left(P \propto \frac{1}{V}\right).$$

As such, Boyle's law can be represented in equation form as:

$$PV = k$$

Therefore, a plot of P against $\frac{1}{V}$ would yield a straight line.

This also means that the product of the pressure and volume of an ideal gas at a certain temperature is constant, and for two conditions of pressure and volume for a sample of gas at constant temperature, we can write the equation shown in

Table 1.

Table 1. Equation for two conditions of pressure and volume for a sample of gas at constant temperature.

Equation	Symbols	Units
$P_1V_1 = P_2V_2$	P_1 = pressure	kiloPascals (kPa)
	P_2 = pressure	kiloPascals (kPa)
	V_1 = volume	cubic decimetres (dm ³)
	V_2 = volume	cubic decimetres (dm ³)

Worked example 3

A 2.20 dm³ gas sample at a pressure of 100. kPa is compressed to 80.0 kPa at constant temperature. Calculate its new volume.

Solution steps

Step 1: Write out the values given in the question.

Step 2: Write out the equation.

Solution steps**Step 3:** Rearrange the equation to make V_2 the subject.**Step 4:** Substitute the values given.**Step 5:** State the answer with appropriate units and the number of significant figures used in rounding.

Relationship between volume and temperature

Charles' law, named after the scientist Jacques Charles, states that at constant pressure the volume of a fixed mass of an ideal gas is directly proportional to its absolute temperature in kelvin ($V \propto T$). This means that if the absolute temperature of a fixed mass of an ideal gas is doubled, the volume of the gas will also double. Charles' law can be represented in equation form as $\frac{V}{T} = k$.

Therefore, the ratio of volume to temperature is constant, and for two conditions of temperature and volume for a sample of gas at constant pressure, we can write the equation shown in **Table 2**.

Table 2. Equation for two conditions of temperature and volume for a sample of gas at constant pressure.

Equation	Symbols	Units
$\frac{V_1}{T_1} = \frac{V_2}{T_2}$	$V_1 = \text{volume}$	cubic decimetres (dm ³)
	$V_2 = \text{volume}$	cubic decimetres (dm ³)
	$T_1 = \text{temperature}$	kelvin (K)
	$T_2 = \text{temperature}$	kelvin (K)

This is what happens in car or bicycle tyres on hot days: as the temperature of the air inside the tyres increases, the volume of the air increases, inflating the tyres!

Worked example 4

A 2.20 dm^3 gas sample at 256 K is cooled down to a temperature of 100 K while keeping its pressure constant. Calculate its new volume.

Solution steps

Step 1: Write out the values given in the question.

Step 2: Write out the equation.

Step 3: Rearrange the equation to make V_2 the subject.

Step 4: Substitute the values given.

Step 5: State the answer with appropriate units and the number of significant figures used in rounding.

So what would a graph of volume vs. temperature yield? What do you expect the graph to look like? Since volume and temperature are directly proportional, then the graph will be a straight line. Charles' law is illustrated graphically in **Figure 3**.

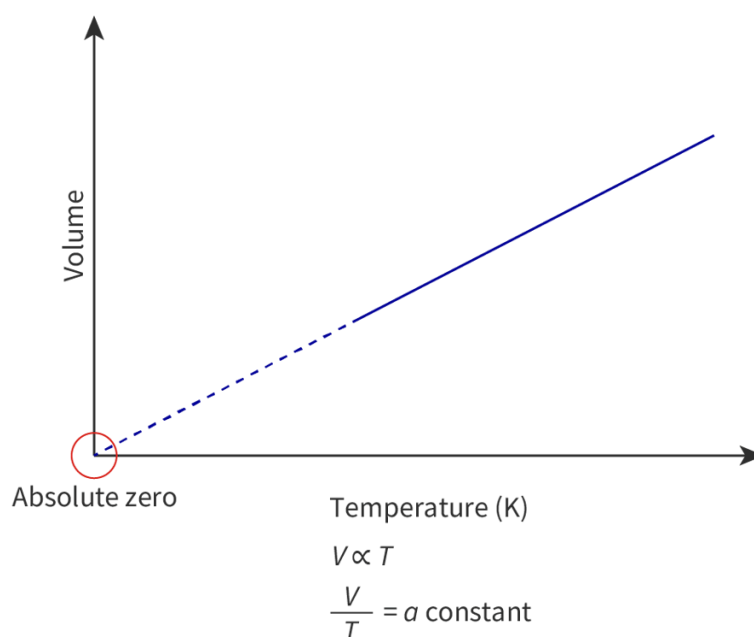


Figure 3. At constant pressure, the volume of a gas is directly proportional to its absolute temperature (in K). The graph can be extrapolated back to absolute zero — the temperature at which a gas would have zero volume.

 More information for figure 3

The image is a graph illustrating the relationship between volume and temperature for a gas. The X-axis represents temperature in Kelvin, ranging from absolute zero upwards. The Y-axis represents volume. This graph demonstrates Charles' Law, showing a direct proportionality between volume and temperature. The graph features a straight, diagonal line starting from the origin (predicting zero volume at absolute zero temperature), extending positively towards the upper right of the graph. Alongside the graph, it has additional text: ' $V \propto T$ ' and ' $V/T = \text{a constant}$ ', which describe the law mathematically. This visual representation highlights the linear relationship described by Charles' Law, indicating that as temperature increases, the volume also increases proportionally, provided the pressure remains constant.

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Worked example 5

A sample of gas occupies 2.20 dm^3 at 30.0°C . Calculate the new volume of this sample when the temperature of this gas is decreased to 15.0°C while keeping its pressure constant.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$V_1 = 2.20 \text{ dm}^3$ $T_1 = 30.0^\circ\text{C}$ $T_2 = 15.0^\circ\text{C}$
Step 2: Write out the equation.	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
Step 3: Rearrange the equation to make V_2 the subject.	$V_2 = \frac{V_1 T_2}{T_1}$
Step 4: Substitute the values given.	In order to use the formula, they must be in Kelvin (you will get the incorrect answer if you use temperature in Celsius). To convert temperature from Celsius to Kelvin, add 273.15 to the Celsius temperature.

Solution steps	Calculations
	$T_1 = 30.0\text{ }^{\circ}\text{C} + 273$ $= 303\text{ K}$ $T_2 = 15.0\text{ }^{\circ}\text{C} + 273$ $= 288\text{ K}$ $V_2 = \frac{(2.20\text{ dm}^3)(288\text{ K})}{(303\text{ K})}$ $V_2 = 2.09\text{ dm}^3$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$V_2 = 2.09\text{ dm}^3$

Relationship between pressure and temperature

Gay-Lussac's law, named after Joseph Louis Gay-Lussac, states that at constant volume the pressure of a fixed mass of an ideal gas is directly proportional to its absolute temperature in kelvin (K). This means that if the absolute temperature of a fixed mass of an ideal gas is doubled, the pressure of the gas will also double.

Gay-Lussac's law can be represented in equation form as $\frac{P}{T} = k$.

Therefore, the ratio of pressure to temperature is constant, and for two conditions of temperature and pressure for a sample of gas at constant volume, we can write the equation shown in **Table 3**. Just as before in Charles' law, the temperature must be expressed in units of kelvin to reflect the absolute temperature of a gas, which would exert no pressure at 0 K, absolute zero.

Table 3. Equation for two conditions of temperature and pressure for a sample of gas at constant volume.

Equation	Symbols	Units
$\frac{P_1}{T_1} = \frac{P_2}{T_2}$	P_1 = pressure	kiloPascals (kPa)
	P_2 = pressure	kiloPascals (kPa)
	T_1 = temperature	kelvin (K)

Equation	Symbols	Units
	$T_1 = \text{temperature}$	kelvin (K)

This relationship can be visualised graphically in **Figure 4**.

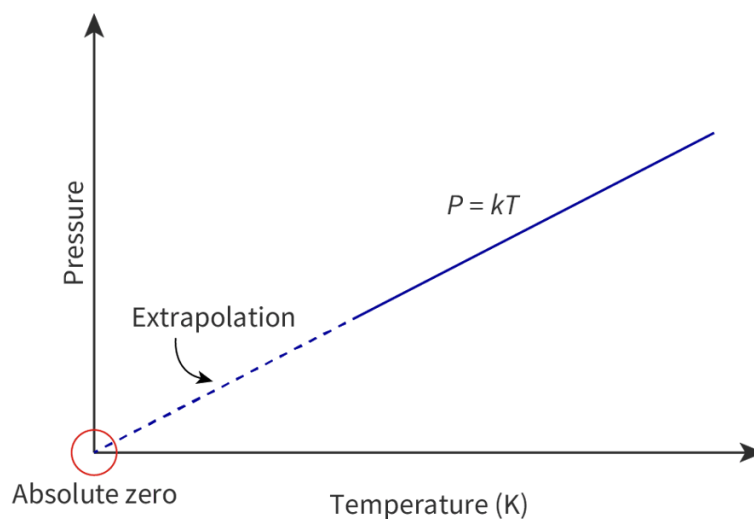


Figure 4. Gay-Lussac's law graphical illustration.

 More information for figure 4

The graph illustrates Gay-Lussac's law, showing the relationship between pressure and temperature. The X-axis represents temperature in Kelvin, starting from absolute zero and extending to higher temperatures. The Y-axis represents pressure, with a linear upward trend indicating a proportional relationship, described by the equation $P = kT$. An extrapolation is indicated, showing the line extending back to theoretically intersect the temperature axis at absolute zero, suggesting that pressure is zero at this point. A solid blue line depicts this relationship with a dashed extension to highlight the extrapolation.

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When a deodorant can is thrown in fire, it explodes! Why? The gas inside heats up, but the volume of the container cannot expand as it is rather rigid. As such, the pressure of the gas increases to an extent that it makes the walls of the can explode.

Worked example 6

A gas sample at 256 K and pressure of 100 kPa is cooled down to a temperature of 100 K while keeping its volume constant. Calculate its new pressure.

Solution steps

Step 1: Write out the values given in the question.

Step 2: Write out the equation.

Step 3: Rearrange the equation to make P_2 the subject.

Step 4: Substitute the values given.

Step 5: State the answer with appropriate units and the number of significant figures used in rounding.

Drag and drop the numbers to make each statement true.

Check your understanding! Drag the correct words into the blanks of the corresponding statements.

The volume of a gas changes by a factor of _____ if the temperature is halved at constant pressure.

The volume of a gas changes by a factor of _____ if the pressure is halved at constant temperature.

The pressure of a gas changes by a factor of _____ if the temperature is quadrupled at constant volume.

The volume of a gas changes by a factor of _____ if both the temperature and pressure are halved.

1 1/2 2 4

✓ Check

Interactive 1. Understanding Gas Behavior: Key Terms.

Activity

- **IB learner profile attribute:** Open-minded
- **Approaches to learning:**
 - Thinking skills
 - Being curious about the natural world
 - Asking questions
 - Applying key ideas and facts in new contexts
 - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 20 minutes
- **Activity type:** Pairs activity

Instructions:

In pairs, you will explore relationships between T , P and V .

1. Open the simulation:

<https://teachchemistry.org/classroom-resources/the-gas-laws-simulation> 
(<https://teachchemistry.org/classroom-resources/the-gas-laws-simulation>)n (<https://teachchemistry.org/classroom-resources/the-gas-laws-simulation>)

2. Press on the tab: Boyle's law. Change volume and press calculate, then press on add data. Repeat until you have a clear graph of the relationship. Describe the relationship. What do you notice about temperature?
3. Reset and repeat step two by changing pressure this time.
4. Press on the tab: Charles' law. Change volume and press calculate, then press on add data. Repeat until you have a clear graph of the relationship. Describe the relationship. What do you notice about pressure?
5. Reset and repeat step two by changing the temperature this time.
6. Press on the tab: Gay-Lussac's law. Change pressure and press calculate, then press on add data. Repeat until you have a clear graph of the relationship. Describe the relationship. What do you notice about volume?
7. Reset and repeat step two by changing the temperature this time.

Ideal gas equation

Learning outcomes

By the end of this section you should be able to solve problems relating to the ideal gas equation.

Can samples of gas with different quantities of particles occupy the same space? Can we calculate the volume by measuring mass, temperature and pressure, using theoretical calculations in advance for practical applications? In this section, we explore the mathematical equations that relate pressure, volume, temperature and amount of moles of an ideal gas.

Practical skills

Tool 3: Mathematics — Using units, symbols and numerical values

In this section, you will be using mathematical tools in carrying out calculations with the ideal gas equation. An important aspect will be that of using the correct SI units in all calculations and being able to round all answers to the correct number of significant figures.

Combined gas law

We have seen that volume is proportional to temperature, volume is inversely proportional to pressure, and pressure is directly proportional to temperature. We were able to write the following relationships:

$$P_1V_1 = P_2V_2$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

In these relationships, 1 and 2 are two different conditions of temperature, pressure and volume of a sample of gas.

These relationships can all be combined into one expression, the combined gas law, in **Table 1**.

Equation	Symbols
$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$	P_1 = pressure
	P_2 = pressure
	V_1 = volume
	V_2 = volume
	T_1 = temperature
	T_2 = temperature

Study skills

When using the combined gas law $\left(\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}\right)$ you need to be careful with the units you use. Temperature should be in kelvin (K), but the units of pressure and volume can be any, as long as they are the same for both sides of the equation.

To convert from °C to K, you should add 273.15; for example, 25 °C is equal to 298.15 K.

Worked example 1

When the absolute temperature of a sample of an ideal gas at 27 °C is doubled the volume increases from 3.00 dm³ to 6.00 dm³. What will be the new temperature of the gas (in °C) if the pressure remains constant?

Solution steps

Step 1: Write out the values given in the question.

Step 2: Write out the equation.

Solution steps
Step 3: Rearrange the equation to make T_2 the subject.
Step 4: Substitute the values given.
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.

Worked example 2

A 2.20 dm^3 gas sample at 256 K and pressure of 100 kPa is cooled down to a temperature of 100 K while compressing it to 80.0 kPa . Calculate the new volume of the gas.

Solution steps	C
Step 1: Write out the values given in the question.	V P T P T
Step 2: Write out the equation.	$\frac{P}{T}$
Step 3: Rearrange the equation to make V_2 the subject.	V
Step 4: Substitute the values given.	V
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	V

Ideal gas equation

The combined gas law suggests that the ratio of the product of pressure and volume to the temperature is equal to a constant:

$$\frac{PV}{T} = \text{constant}$$

We also saw that volume and moles of gas are directly proportional, and therefore the ratio of volume to moles is equal to a constant:

$$\frac{V}{n} = \text{constant}$$

The gas laws can be combined with [Avogadro's law](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/) (see [section S1.4.6](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/>)) to give the ideal gas equation shown in **Table 2**.

Table 2. Ideal gas equation.

Equation	Symbols	Units
$PV = nRT$	P is the pressure	pascals (Pa)
	V is the volume	cubic metres (m ³)
	n is the amount of gas	moles
	R is the universal gas constant (8.31 J K ⁻¹ mol ⁻¹)	joules per kelvin per mole (J K ⁻¹ mol ⁻¹)
	T is the absolute temperature	kelvin (K)

In this equation, the relationships discussed in [section S1.5.3](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-volume-of-an-ideal-gas-and-gas-laws-id-44352/>) are represented mathematically.

Study skills

The value of the gas constant R , the ideal gas equation, and the combined gas law, are given in [section 1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/>) and [section 2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/>) in the DP chemistry data booklet.

When using the ideal gas equation, be careful to use the correct units:

- Pressure, P , must be in Pa
- Volume, V , must be in m^3
- Amount in mol, n , this can be calculated using $n = \frac{m}{M}$, where m is the mass in grams and M is the molar mass.
- Temperature, T , must be converted to kelvin (K); if $^{\circ}\text{C}$ is given, add 273.15

It should also be noted that it is possible to use kPa for the pressure if the volume is given in dm^3 .

The ideal gas equation can be rearranged to calculate the amount (in mol) of a gas (n) in **Table 3**.

Table 3. Equation for the amount of a gas

Equation	Symbols	Units
$n = \frac{PV}{RT}$	n = amount of gas	moles (mol)
	P = pressure	pascals (Pa)
	V = volume	cubic metres (m^3)
	T = temperature	kelvin (K)
	R = universal gas constant ($8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)	joules per kelvin per mol ($\text{J K}^{-1} \text{ mol}^{-1}$)

Worked example 3

Calculate the moles and mass in grams of carbon dioxide that occupies 20.0 cm^3 at 25.0°C and 100.0 kPa .

Solution steps	Calc
Step 1: Write out the values given in the question.	$V =$ $T =$ $P =$

Solution steps	Calc
Step 2 Convert the volume to m ³ and pressure to Pa.	$\frac{4}{100}$ $P =$
Step 3: Write out the equation.	$n =$
Step 4: Substitute the values given.	$n =$
Step 5: Substitute the values given.	$n =$
Step 6: Write out the equation to convert moles to mass.	$m =$
Step 7: Substitute the values.	$m =$
Step 8: State the answer with appropriate units and the number of significant figures used in rounding.	$m =$

What if you didn't know the identity of a gas but you were able to make a measurement of mass? Then you can calculate molar mass (M).

In subtopic S1.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/>), we have learned that:

$$n \text{ (in moles)} = \frac{m \text{ (g)}}{M \text{ (g mol}^{-1}\text{)}}$$

Substituting n with $\frac{m}{M}$, we obtain:

$$\frac{m}{M} = \frac{PV}{RT}$$

Rearranging the equation for M , we obtain the equation for molar mass shown in **Table 4**.

Table 4. Equation for molar mass.

Equation	Symbols	Units
$M = \frac{mRT}{PV}$	M = molar mass	grams per mole (g mol ⁻¹)
	m = mass of gas	grams (g)
	P = pressure	pascals (Pa)
	V = volume	cubic metres (m ³)
	T = temperature	kelvin (K)
	R = universal gas constant (8.31 J K ⁻¹ mol ⁻¹)	joules per kelvin per mole (J K ⁻¹ mol ⁻¹)

Worked example 4

Calculate the molar mass of a gas if a 500.0 cm³ sample at 20.0 °C and 101.325 kPa has a mass of 0.666 g.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$V = 500.0 \text{ cm}^3$ $T = 293.15 \text{ K}$ $P = 101.325 \text{ kPa}$ $m = 0.666 \text{ g}$
Step 2: Convert the volume to m ³ and pressure to Pa.	$V = \frac{500}{1000000} \text{ m}^3$ $P = 101325 \text{ Pa}$
Step 3: Write the equation.	$M = \frac{m}{n}$
Step 4: Substitute the values given.	$M = \frac{0.666}{n}$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$M = 32 \text{ g mol}^{-1}$

Practical skills

Tool 3: Mathematics — Using units, symbols and numerical values

In this section, you are applying mathematical tools to determine variables like (mass and volume) from data as accurately as possible.

International Mindedness

Units are not the same worldwide. For example, the Pascal (Pa) is the SI unit for pressure; nonetheless, several other units continue to be widely used worldwide by both scientists and non-scientists. They include atmosphere (atm), mm Hg (millimetres of mercury), Torr, bar, and psi (pounds per square inch). Due to its close proximity to 1 atm and the fact that many weather maps display pressure in millibars, the bar (1.00×10^5 Pa) is currently an often used unit. Similarly, even though the litre (dm^3) is far more frequently used, the SI unit for volume is the m^3 . This is why in science we use the SI system of units. It allows universal access to science understanding.

Practical use of the ideal gas equation:

The molar mass of a gas such as butane (C_4H_{10}) can be determined experimentally using the ideal gas equation. In the experimental procedure shown in **Figure 1**, a known volume of gas from a cigarette lighter is collected in a gas syringe. The change in mass of the lighter is also recorded which gives the mass of gas collected. Together with the pressure and the temperature, the molar mass of the gas can be determined.



Figure 1. Collecting butane gas in a gas syringe.

Worked example 5

50.0 cm³ of gas from a lighter was collected in a gas syringe. The change in mass of the lighter was recorded as 0.115 g. The temperature and atmospheric pressure at which the gas was collected are 298 K and 96.931 kPa respectively. Using this data, determine the molar mass of the gas.

Solution steps	Calculations
Step 1: Write out the values given in the question.	m is the change in mass (in g) = 0.115 g R is the universal gas constant (8.314 J K⁻¹ mol⁻¹) P is the pressure of the measuring cylinder = V is the volume of gas in measuring cylinder 50
Step 2: Write out the equation.	$M = \frac{mRT}{PV}$
Step 3: Substitute the values given.	$M = \frac{0.115 \times 8.31 \times 298}{96931 \times 0.000050}$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$M = 58.8 \text{ g mol}^{-1}$

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:**
 - Thinking skills
 - Being curious about the natural world
 - Asking questions and framing hypotheses based upon sensible scientific rationale
 - Designing procedures and models

- Applying key ideas and facts in new contexts
- Engaging with, and designing linking questions
 - Social skills —Working collaboratively to achieve a common goal
- **Time required to complete activity:** 20 minutes.
- **Activity type:** Group activity

Instructions:

In a group of 2—3:

1. Find an unused box or container in your house or classroom, like a shoe box.
2. Use a ruler to record measurements that will help you calculate the volume.
3. Consult a local weather source to determine the air pressure and the daily high and low temperatures for that day.
4. Calculate the number of moles and molecules of oxygen that could fit in the box at the warmest and coldest part of the day.

Answer the following questions:

- What variables do you need to determine?
- How can you obtain them?
- What calculations do you need to perform?
- Which unit conversions do you need to apply?

Do not forget to list all resources used.